Xin Huey Wong

+6017-272-9230 | xinhuey.wong@gmail.com | linkedin.com/in/xhwong | github.com/xinhuey | Website

Education

University of Waterloo Waterloo, Canada

Bachelor of Computer Science, Software Engineering Specialisation *Sept 2021 – Apr 2026*

- Relevant Coursework: Algorithms, Data Structures, Intro to AI, Software Design & Architecture, Testing & QA
  CGPA: 72.84 (First Class Honours)

Experience

Full Stack Developer Intern Sept 2025 - Dec 2025

Compass Digital *Toronto, Canada*

- Improved **platform content maintainability**, as measured by reduced manual update time for client-facing cafeteria applications, by building a dynamic announcements module in **Vue.js** with configurable embedded links.
- Developed an AI marketing agent prototype that used real-time user behaviour and order data to generate personalised promotion strategies, **boosting user engagement and outreach by 80%.**
- Implemented automated API test cleanup and end-to-end tests for a third-party rewards integration, **reducing failed CI runs by 70%**.

Software Engineering Intern Jan 2025 - Apr 2025

Cactus Information Systems *Malaysia*

- Designed and implemented WPF desktop application features using C# and **MVVM** principles for a production line information system used by **100+ daily operators**, decoupling UI and business logic across 30 modules.
- Built interactive forms with real-time validation and MySQL integration, **reducing manual data-entry errors by 98%** across supplier and customer workflows.

Software Developer Intern May 2024 - Aug 2024

Imagine Communications *Waterloo, Canada*

- Developed **Python** and **Selenium** automated testing frameworks, cutting regression testing time by **50%** and embedding testing into the **Agile development cycle.**
- Maintained **Docker containers and Kubernetes pods** for the Versio Control broadcasting application, **supporting 80% uptime**.
- Configured AWS EC2 AMIs, **reducing instance deployment time by 85%** across 3 environments.

VP of Development, UW Data Science Club Sept 2024 - Dec 2024

University of Waterloo *Waterloo, Canada*

- Led a team of 5 student developers to build a **Next.js and MongoDB** QR code check-in platform with REST APIs, streamlining event registration for **100+ participants.**
- Built Python scripts to validate and reconcile data across MongoDB and CSV sources, **catching 5 data discrepancies** per event cycle.
- Scaled and maintained a member-facing website serving **300+ users**, ensuring consistent uptime and performance.

Software Engineering Intern Sept 2023 - Dec 2023

Christie Digital Systems *Kitchener, Canada*

- Built and debugged CAWS, a Python/Selenium web UI test automation framework, **raising code coverage to 95%** and catching 5 critical defects ahead of the MicroTiles LED software release.
- Engineered a live log-monitoring tool in Python (PyQt5) that opens telnet sessions and streams logs into a structured table view, **cutting testing turnaround time by 40%**, and developed a Python utility to automate EDID file format conversion (.bin to .txt).

Projects

PerryVision | *Next.js, FastAPI, PyTorch, TypeScript*

- Built a full-stack binary image classifier with **decoupled inference, API, and client layers**; **fine-tuned ResNet18 via transfer learning**, achieving **97% test accuracy** on a self-collected dataset.

TriModel Options Engine | *Python*

- Built a Python options pricing engine (**Black-Scholes, Monte Carlo, Binomial Tree**) with **37 automated tests** validating cross-model convergence, Greeks, and no-arbitrage identities.

Technical Skills

Languages:Python, C/C++, C#, JavaScript, SQL (PostgreSQL, MySQL)
Frontend: React.js, Vue.js, HTML/CSS, Chakra UI
Backend: Node.js, Express.js, Flask, REST APIs, MongoDB
DevOps & Cloud: Docker, Kubernetes, AWS (EC2), CI/CD, Git/GitHub
Testing: Selenium, Playwright, Postman, Unit Testing, End-to-End Testing